

HIGH PERFORMANCE COMPUTING FOR SCIENCE AND ENGINEERING

Harvard University
CS 2050

Lecture 23

April 28, 2026

QUIZ 12



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2025-2026 Spring

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Assign To



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Quiz 12

The quizzes are designed to take no more than 30 minutes, but one hour will be allowed.

Do not click into the quiz until you are ready to take it.

Quiz Type Graded Quiz

Points 10

Assignment Group Quizzes

Shuffle Answers No

Time Limit 60 Minutes

FINAL EXAM

 CS-2050 / final-exam 

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 README  

Final Exam

 Important

- CS 2050: In-person exam, **Wed May 13, 9am-noon, Science Center, Hall C.**
- CSCI E-205: Proctorio exam, opening **Wed May 13, 9am** and closing after 24 hours. Students must complete the exam within three hours of opening it.

FINAL EXAM

- CS 2050: In-person exam, **Wed May 13, 9am-noon, Science Center, Hall C.**
- CSCI E-205: Proctorio exam, opening **Wed May 13, 9am** and closing after 24 hours. Students must complete the exam within three hours of opening it.

What to expect

- The exam is designed to measure students' engagement with the course over the full semester. Unless specifically excluded, any material from the lectures, quizzes, and homework assignments is admissible. **However, we suggest focusing your studying on the topics listed below, as well as the quizzes and reading assignments.**
- The exam format will be similar to the quiz format, consisting mostly of multiple choice and short answer questions. However, unlike the quizzes, the exam is fully "closed book," meaning no references (paper or electronic) are permitted. For this reason, the exam will be more difficult than the quizzes.
- We will not require detailed mathematical manipulations.

WEEK 1

- Context for the course
- Moore's law and other computing trends
- Parallel computing basics
- Flynn's taxonomy
- Basic knowledge of the course compute cluster
- Git and GitHub basics
- Slurm basics
- CMake basics
- pybind11 basics

WEEK 2

- Hardware: basic components
- Cache memory
- Typical latencies, and latency vs throughput
- Memory access patterns and implications
- SIMD vectorization
- GPU basics

WEEK 3

- Shared memory parallelism
- Processes and threads
- Race conditions and atomic operations
- OpenMP basics
- Reductions

WEEK 4

- Distributed memory parallelism
- MPI basics
- Point-to-point and collective communication
- Blocking vs non-blocking operations
- Amdahl's law

WEEK 5

- CPU vs GPU fundamentals
- High-level understanding of the Thrust library (no detailed syntax)
- Host memory vs device memory (including `thrust::universal_vector`, `thrust::host_vector`, and `thrust::device_vector`)
- High-level understanding of CuPy and Numba CUDA (no detailed syntax)

WEEK 6

- VTune basics
- Strong and weak scaling
- CUDA basics
- Fundamentals of Nsight Systems and Nsight Compute

WEEK 8

- High-level understanding of LLM training and opportunities for parallelization

WEEK 9

- BLAS and LAPACK basics (particularly if appearing in homework assignments)
- High-level understanding of distributed linear algebra
- High-level aims of the Julia language

WEEK 10

- High-level understanding of molecular dynamics and opportunities for parallelization

WEEK 11

- High-level aims for the Kokkos ecosystem
- Roofline plots

RECALL FROM WEEK 1

- Are you comfortable using the Linux command line?
- Are you comfortable using Bash scripts?
- Are you comfortable using C++?
- Are you comfortable using Python?
- Are you comfortable using Git and GitHub?
- Are you comfortable using CMake?
- Are you comfortable using Slurm?
- Are you comfortable using VS Code?